

## YEAR 7 MATHEMATICS · CHAPTER 4

# Fractions and percentages

- |                                       |                                             |
|---------------------------------------|---------------------------------------------|
| 4.01 Fractions                        | 4.02 Ordering fractions                     |
| 4.03 Adding and subtracting fractions | 4.04 Adding and subtracting mixed numerals  |
| 4.05 Fraction of a quantity           | 4.06 Multiplying fractions                  |
| 4.07 Dividing fractions               | 4.08 Fraction problems                      |
| 4.09 Fractions on a calculator        | 4.10 Percentages, fractions and decimals    |
| 4.11 Percentage of a quantity         | 4.12 Expressing as a fraction or percentage |

Name \_\_\_\_\_

Class \_\_\_\_\_

Teacher \_\_\_\_\_

## How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

### WE DO

**Grey boxes with a dark border.** Work with your teacher. Copy what they write on the board.

### YOU DO

**Blue boxes.** Work on your own. Show your working in the box and write your answer.

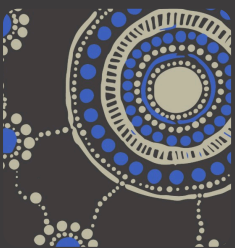
## Contents

| Lesson | Topic                                  | Page | Done                     |
|--------|----------------------------------------|------|--------------------------|
| 4.01   | Fractions                              | 3    | <input type="checkbox"/> |
| 4.02   | Ordering fractions                     | 7    | <input type="checkbox"/> |
| 4.03   | Adding and subtracting fractions       | 11   | <input type="checkbox"/> |
| 4.04   | Adding and subtracting mixed numerals  | 15   | <input type="checkbox"/> |
| 4.05   | Fraction of a quantity                 | 19   | <input type="checkbox"/> |
| 4.06   | Multiplying fractions                  | 23   | <input type="checkbox"/> |
| 4.07   | Dividing fractions                     | 27   | <input type="checkbox"/> |
| 4.08   | Fraction problems                      | 31   | <input type="checkbox"/> |
| 4.09   | Fractions on a calculator              | 35   | <input type="checkbox"/> |
| 4.10   | Percentages, fractions and decimals    | 39   | <input type="checkbox"/> |
| 4.11   | Percentage of a quantity               | 43   | <input type="checkbox"/> |
| 4.12   | Expressing as a fraction or percentage | 47   | <input type="checkbox"/> |

## In this chapter you will

- write equivalent fractions and simplify
- order fractions
- add, subtract, multiply and divide fractions and mixed numerals
- find a fraction or percentage of a quantity
- solve fraction problems
- convert between fractions, decimals and percentages

NSW Mathematics K-10 Syllabus (2022), Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.



### Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON  
4.01

## Fractions

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- convert between mixed numerals and improper fractions
- write equivalent fractions and simplify fractions

## Success criteria

I can:

- ☐ change a mixed numeral to an improper fraction and back
- ☐ multiply top and bottom by the same number to find an equivalent fraction
- ☐ divide top and bottom by the HCF to simplify

**Key terms** Write the definition as your teacher explains it.**Improper fraction****Mixed numeral****Equivalent**

Notes

## Level 1 · Getting started

**WE DO**

With your teacher: copy what they write on the board.

**Example 1**Write  $1\frac{3}{4}$  as an improper fraction.**Example 2**Write  $\frac{7}{4}$  as a mixed numeral.**YOU DO**

On your own: Set A. Write each mixed numeral as an improper fraction.

①  $1\frac{3}{4}$

④  $3\frac{1}{7}$

②  $2\frac{4}{9}$

⑤  $4\frac{1}{2}$

③  $5\frac{3}{5}$

⑥  $2\frac{5}{6}$

**YOU DO**

On your own: Set B. Write each improper fraction as a mixed numeral.

1  $\frac{7}{4}$

4  $\frac{11}{2}$

2  $\frac{26}{3}$

5  $\frac{17}{5}$

3  $\frac{23}{7}$

6  $\frac{36}{24}$

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

**Example 3**

Complete:  $\frac{1}{3} = \frac{?}{15}$

**Example 4**

Complete:  $\frac{3}{4} = \frac{12}{?}$

**YOU DO**

On your own: Set C. Complete each pair of equivalent fractions.

1  $\frac{1}{3} = \frac{?}{15}$

3  $\frac{4}{5} = \frac{?}{20}$

5  $\frac{1}{2} = \frac{?}{50}$

2  $\frac{3}{4} = \frac{12}{?}$

4  $\frac{5}{12} = \frac{15}{?}$

6  $\frac{3}{20} = \frac{?}{100}$



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Simplify  $\frac{25}{30}$ .
**Example 6** Simplify  $\frac{36}{24}$  and write as a mixed numeral.
**YOU DO** On your own: Set D. Simplify each fraction.1  $\frac{25}{30}$ 

3  $\frac{15}{20}$ 

5  $\frac{24}{48}$ 

2  $\frac{30}{40}$ 

4  $\frac{20}{15}$ 

6  $\frac{24}{42}$ 


✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**Write  $3\frac{2}{5}$  as an improper fraction.
**Level 2**Complete:  $\frac{2}{7} = \frac{?}{35}$ 

**Level 3**Simplify  $\frac{45}{60}$ .

## Extension · Stretch yourself

### Worked example

Find three fractions equivalent to  $\frac{2}{3}$  with denominators between 10 and 20.

1. Multiply top and bottom by the same number
2.  $\times 4, \times 5, \times 6$

Answer:  $\frac{8}{12}, \frac{10}{15}, \frac{12}{18}$

### YOU DO

On your own: read the worked example, then try these.

- E1 Which fraction is not equivalent to  $\frac{3}{5}$ :  $\frac{9}{15}, \frac{12}{20}, \frac{15}{30}, \frac{30}{50}$ ?

- E2 A fraction simplifies to  $\frac{2}{7}$ . Its numerator and denominator add to 45. Find the fraction.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Mixed  $\rightarrow$  improper: whole  $\times$  bottom + top.
  2. Improper  $\rightarrow$  mixed: divide top by bottom.
  3. Equivalent:  $\times$  or  $\div$  top and bottom by the same number.
  4. Simplify:  $\div$  top and bottom by the HCF.

### Level 1

$$1\frac{3}{4}$$

1.  $1 \times 4 + 3 = 7$

Answer:  $\frac{7}{4}$

### Level 2

$$\frac{26}{3}$$

1.  $26 \div 3 = 8 \text{ r } 2$

Answer:  $8\frac{2}{3}$

### Level 3

$$\text{Simplify } \frac{24}{42}$$

1. HCF is 6

Answer:  $\frac{4}{7}$

### Extension

$$\text{Equivalent to } \frac{2}{3}$$

1.  $\times 4, \times 5, \times 6$

Answer:  $\frac{8}{12}, \frac{10}{15}, \frac{12}{18}$

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LESSON  
4.02

## Ordering fractions

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- compare and order fractions

## Success criteria

I can:

- ☐ compare fractions with the same denominator
- ☐ change fractions to a common denominator
- ☐ order fractions from smallest to largest

**Key terms** Write the definition as your teacher explains it.**Common denominator****Ascending order**

Notes

## Level 1 · Getting started

**WE DO** With your teacher: copy what they write on the board.**Example 1** Which is larger:  $\frac{3}{8}$  or  $\frac{5}{8}$ ?**Example 2** Which is larger:  $\frac{2}{3}$  or  $\frac{3}{5}$ ?**YOU DO** On your own: Set A. Which fraction is larger?1  $\frac{3}{7}$  or  $\frac{5}{7}$ 4  $\frac{1}{2}$  or  $\frac{2}{5}$ 2  $\frac{1}{4}$  or  $\frac{1}{3}$ 5  $\frac{2}{3}$  or  $\frac{5}{6}$ 3  $\frac{9}{10}$  or  $\frac{7}{10}$ 6  $\frac{3}{4}$  or  $\frac{5}{8}$

**YOU DO** On your own: Set B. Write each fraction with the denominator shown.

1  $\frac{1}{2} \rightarrow$  twelfths

4  $\frac{2}{5} \rightarrow$  tenths

2  $\frac{2}{3} \rightarrow$  twelfths

5  $\frac{5}{6} \rightarrow$  twelfths

3  $\frac{3}{4} \rightarrow$  twelfths

6  $\frac{3}{8} \rightarrow$  twenty-fourths

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Order  $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$  from smallest to largest.**Example 4** Order  $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$ .**YOU DO** On your own: Set C. Order from smallest to largest.

1  $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$

3  $\frac{5}{8}, \frac{1}{2}, \frac{3}{4}, \frac{3}{8}$

2  $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$

4  $\frac{2}{3}, \frac{4}{9}, \frac{5}{6}, \frac{1}{2}$



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Order  $\frac{3}{4}$ ,  $\frac{2}{3}$ ,  $\frac{5}{6}$ ,  $\frac{7}{12}$ .**Example 6** Order  $1\frac{1}{3}$ ,  $\frac{5}{4}$ ,  $\frac{11}{8}$ .
**YOU DO** On your own: Set D. Order from smallest to largest.**1**  $\frac{3}{4}$ ,  $\frac{2}{3}$ ,  $\frac{5}{6}$ ,  $\frac{7}{12}$ **3**  $1\frac{1}{3}$ ,  $\frac{5}{4}$ ,  $\frac{11}{8}$ ,  $1\frac{1}{6}$ **2**  $\frac{3}{5}$ ,  $\frac{1}{2}$ ,  $\frac{7}{10}$ ,  $\frac{4}{15}$ **4** Find a fraction between  $\frac{1}{3}$  and  $\frac{1}{2}$ .

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**Which is larger:  $\frac{4}{9}$  or  $\frac{7}{9}$ ?
**Level 2**Which is larger:  $\frac{3}{4}$  or  $\frac{4}{5}$ ?
**Level 3**Order  $\frac{5}{6}$ ,  $\frac{2}{3}$ ,  $\frac{3}{4}$ .

## Extension · Stretch yourself

### Worked example

Order  $\frac{5}{7}$ ,  $\frac{2}{3}$ ,  $\frac{7}{10}$  from smallest to largest.

1. Common denominator 210

2.  $\frac{150}{210}$ ,  $\frac{140}{210}$ ,  $\frac{147}{210}$

Answer:  $\frac{2}{3}$ ,  $\frac{7}{10}$ ,  $\frac{5}{7}$

### YOU DO

On your own: read the worked example, then try these.

E1 Order  $\frac{4}{9}$ ,  $\frac{3}{7}$ ,  $\frac{1}{2}$ .

E2 Find a fraction exactly halfway between  $\frac{1}{4}$  and  $\frac{1}{3}$ .

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Same denominator: compare the numerators.
  2. Different denominators: find the LCM of the denominators.
  3. Rewrite each fraction with that denominator.
  4. Order the numerators.

### Level 1

$\frac{3}{8}$  or  $\frac{5}{8}$

1. Same denominator

Answer:  $\frac{5}{8}$

### Level 2

$\frac{2}{3}$  or  $\frac{3}{5}$

1.  $\frac{10}{15}$  vs  $\frac{9}{15}$

Answer:  $\frac{2}{3}$

### Level 3

$\frac{1}{2}$ ,  $\frac{1}{3}$ ,  $\frac{7}{12}$ ,  $\frac{11}{12}$

1. Twelfths: 6, 4, 7, 11

Answer:  $\frac{1}{3}$ ,  $\frac{1}{2}$ ,  $\frac{7}{12}$ ,  $\frac{11}{12}$

### Extension

$\frac{5}{7}$ ,  $\frac{2}{3}$ ,  $\frac{7}{10}$

1. Denominator 210

Answer:  $\frac{2}{3}$ ,  $\frac{7}{10}$ ,  $\frac{5}{7}$

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LESSON  
4.03

## Adding and subtracting fractions

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- add and subtract fractions

## Success criteria

I can:

- ☐ add and subtract fractions with the same denominator
- ☐ use a common denominator when the denominators are different
- ☐ simplify the answer

**Key terms** Write the definition as your teacher explains it.**Numerator****Denominator**

Notes

## Level 1 · Getting started

## WE DO

With your teacher: copy what they write on the board.

## Example 1

$\frac{3}{4} + \frac{1}{4}$

## Example 2

$\frac{5}{7} - \frac{2}{7}$

## YOU DO

On your own: Set A. Evaluate. Simplify if you can.

1  $\frac{3}{4} + \frac{1}{4}$

4  $\frac{5}{8} - \frac{1}{8}$

2  $\frac{2}{9} + \frac{5}{9}$

5  $\frac{4}{5} + \frac{3}{5}$

3  $\frac{7}{10} - \frac{3}{10}$

6  $\frac{11}{12} - \frac{5}{12}$

**YOU DO**

On your own: Set B. Evaluate.

1  $\frac{1}{2} + \frac{1}{4}$

4  $\frac{5}{10} - \frac{1}{3}$

2  $\frac{2}{3} - \frac{1}{6}$

5  $\frac{1}{2} - \frac{3}{8}$

3  $\frac{2}{5} + \frac{3}{10}$

6  $\frac{3}{4} + \frac{1}{8}$

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

**Example 3**

$\frac{2}{5} + \frac{3}{10}$

**Example 4**

$\frac{3}{4} - \frac{1}{5}$

**YOU DO**

On your own: Set C. Evaluate. Use a common denominator.

1  $\frac{1}{2} + \frac{1}{3}$

3  $\frac{2}{3} + \frac{1}{4}$

5  $\frac{5}{6} + \frac{3}{4}$

2  $\frac{3}{4} - \frac{1}{5}$

4  $\frac{4}{5} - \frac{1}{2}$

6  $\frac{13}{18} - \frac{3}{8}$



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.

**Example 5**  $\frac{1}{2} + \frac{1}{3}$

**Example 6**  $\frac{5}{6} + \frac{3}{4}$

**YOU DO** On your own: Set D. Evaluate.

**1**  $\frac{5}{6} + \frac{4}{5}$

**3**  $\frac{7}{8} - \frac{1}{3} - \frac{1}{6}$

**2**  $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$

**4** Tom ate  $\frac{1}{3}$  of a pizza and Sue ate  $\frac{2}{5}$ . What fraction is left?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO**

## Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

$\frac{2}{7} + \frac{3}{7}$

**Level 2**

$\frac{1}{2} - \frac{1}{6}$

**Level 3**

$\frac{2}{3} + \frac{3}{5}$

## Extension · Stretch yourself

### Worked example

Find the missing fraction:  $\frac{3}{8} + ? = \frac{5}{6}$

1.  $? = \frac{5}{6} - \frac{3}{8}$

2.  $= \frac{20}{24} - \frac{9}{24}$

Answer:  $\frac{11}{24}$

### YOU DO

On your own: read the worked example, then try these.

E1 Find the missing fraction:  $? - \frac{2}{5} = \frac{1}{3}$

E2 Write  $\frac{7}{12}$  as the sum of two different unit fractions (numerator 1).

## Lesson summary

Missed the lesson? Start here.

**Remember**

1. Same denominator: add or subtract the numerators.
3. Rewrite, then add or subtract.

2. Different: find the lowest common denominator.
4. Simplify the answer.

### Level 1

$$\frac{3}{4} + \frac{1}{4}$$

1.  $\frac{4}{4}$

Answer: 1

### Level 2

$$\frac{2}{5} + \frac{3}{10}$$

1.  $\frac{4}{10} + \frac{3}{10}$

Answer:  $\frac{7}{10}$

### Level 3

$$\frac{3}{4} - \frac{1}{5}$$

1.  $\frac{15}{20} - \frac{4}{20}$

Answer:  $\frac{11}{20}$

### Extension

$$\frac{3}{8} + ? = \frac{5}{6}$$

1.  $\frac{20}{24} - \frac{9}{24}$

Answer:  $\frac{11}{24}$

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LESSON  
4.04

## Adding and subtracting mixed numerals

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- add and subtract mixed numerals

## Success criteria

I can:

- ☐ add the whole numbers, then the fractions
- ☐ change mixed numerals to improper fractions when subtracting needs it
- ☐ write the answer as a simplified mixed numeral

**Key terms** Write the definition as your teacher explains it.**Mixed numeral****Whole number**

Notes

## Level 1 · Getting started

## WE DO

With your teacher: copy what they write on the board.

## Example 1

$2\frac{1}{5} + 6\frac{1}{2}$

## Example 2

$3\frac{3}{4} + 1\frac{1}{2}$

## YOU DO

On your own: Set A. Evaluate.

①  $2\frac{1}{5} + 1\frac{2}{5}$

④  $5\frac{1}{2} + 3$

②  $3\frac{1}{4} + 2\frac{1}{4}$

⑤  $6\frac{7}{9} - 2\frac{4}{9}$

③  $4\frac{5}{7} - 1\frac{2}{7}$

⑥  $1\frac{1}{3} + 1\frac{2}{3}$

**YOU DO** On your own: Set B. Evaluate.

①  $10 - \frac{1}{3}$

④  $4 - 1\frac{2}{5}$

②  $5 - \frac{3}{4}$

⑤  $3\frac{5}{6} + \frac{1}{6}$

③  $2\frac{3}{4} + 1\frac{3}{4}$

⑥  $2\frac{1}{8} - \frac{3}{8}$

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.

**Example 3**  $10 - \frac{1}{3}$

**Example 4**  $3\frac{2}{3} - 2\frac{1}{6}$

**YOU DO** On your own: Set C. Evaluate.

①  $2\frac{1}{5} + 6\frac{1}{2}$

③  $3\frac{3}{4} + 1\frac{1}{2}$

⑤  $1\frac{2}{3} + 2\frac{3}{4}$

②  $3\frac{2}{3} - 2\frac{1}{6}$

④  $4\frac{3}{4} - 2\frac{1}{3}$

⑥  $3\frac{5}{6} - 1\frac{1}{4}$



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.

**Example 5**  $5\frac{1}{4} - 3\frac{2}{3}$

**Example 6**  $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$

**YOU DO** On your own: Set D. Evaluate.

①  $5\frac{1}{4} - 3\frac{2}{3}$

③  $2\frac{1}{2} + 1\frac{1}{3} + 2\frac{3}{4}$

②  $3\frac{1}{2} - 1\frac{4}{5}$

④ A plank is  $3\frac{1}{2}$  m long.  $1\frac{3}{4}$  m is cut off. How long is the rest?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

● Level 1

$2\frac{1}{3} + 4\frac{1}{3}$

●● Level 2

$6 - \frac{2}{5}$

●●● Level 3

$4\frac{1}{3} - 2\frac{3}{4}$

## Extension · Stretch yourself

### Worked example

Evaluate  $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$ .

1. Improper:  $\frac{11}{4} - \frac{8}{5} + \frac{3}{2}$
2. Denominator 20:  $\frac{55}{20} - \frac{32}{20} + \frac{30}{20}$
3.  $\frac{53}{20}$

Answer:  $2\frac{13}{20}$

### YOU DO

On your own: read the worked example, then try these.

- E1 Evaluate  $5\frac{1}{3} - 2\frac{1}{2} - 1\frac{3}{4}$ .

- E2 Two mixed numerals add to  $7\frac{1}{6}$ . One is  $3\frac{2}{3}$ . Find the other.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Adding: add wholes, then add fractions.
  2. Carry if the fraction is more than 1.
  3. Subtracting: change to improper fractions if needed.
  4. Write the answer as a mixed numeral.

### Level 1

$$2\frac{1}{5} + 6\frac{1}{2}$$

1.  $8 + \frac{7}{10}$

Answer:  $8\frac{7}{10}$

### Level 2

$$10 - \frac{1}{3}$$

1.  $9\frac{3}{3} - \frac{1}{3}$

Answer:  $9\frac{2}{3}$

### Level 3

$$5\frac{1}{4} - 3\frac{2}{3}$$

1.  $\frac{63}{12} - \frac{44}{12}$

Answer:  $1\frac{7}{12}$

### Extension

$$2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$$

1.  $\frac{53}{20}$

Answer:  $2\frac{13}{20}$

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LESSON  
4.05

## Fraction of a quantity

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find a fraction of a quantity, including with units

## Success criteria

I can:

- ☐ divide by the denominator, then multiply by the numerator
- ☐ convert units before finding the fraction
- ☐ give the answer in the unit asked for

**Key terms** Write the definition as your teacher explains it.**Quantity****Unit**

Notes

## Level 1 · Getting started

## WE DO

With your teacher: copy what they write on the board.

## Example 1

 $\frac{1}{4}$  of 28

## Example 2

 $\frac{3}{7}$  of 21

## YOU DO

On your own: Set A. Evaluate.

1  $\frac{1}{4}$  of 28
2  $\frac{1}{10}$  of 35
3  $\frac{2}{5}$  of 20
4  $\frac{4}{9}$  of 27
5  $\frac{2}{3}$  of 30
6  $\frac{3}{7}$  of 21

**YOU DO** On your own: Set B. Evaluate.

1  $\frac{3}{8}$  of 64

4  $\frac{3}{4}$  of 1000

2  $\frac{5}{6}$  of 42

5  $\frac{2}{9}$  of 54

3  $\frac{7}{10}$  of 150

6  $\frac{5}{12}$  of 96

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3**  $\frac{2}{3}$  of 1 day (in hours)**Example 4**  $\frac{4}{5}$  of 2 km (in metres)**YOU DO** On your own: Set C. Evaluate in the unit shown.

1  $\frac{2}{3}$  of 1 day (hours)

3  $\frac{3}{4}$  of 3 h (min)

5  $\frac{1}{6}$  of 5 min (s)

2  $\frac{4}{5}$  of 2 km (m)

4  $\frac{7}{10}$  of 15 cm (mm)

6  $\frac{5}{8}$  of 2 m (cm)



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5**  $\frac{3}{4}$  of 3 hours (in minutes)**Example 6**  $\frac{1}{6}$  of 5 minutes (in seconds)
**YOU DO** On your own: Set D. Word problems.

- 1  $\frac{3}{5}$  of a class of 30 are girls. How many boys are there?

- 3  $\frac{2}{3}$  of a number is 24. Find the number.

- 2 A phone costs \$840. Mia has saved  $\frac{5}{7}$  of the price. How much more does she need?

- 4 Which is more:  $\frac{3}{4}$  of an hour or  $\frac{7}{10}$  of an hour? By how many minutes?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO**

## Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1** $\frac{1}{5}$  of 45
**Level 2** $\frac{5}{6}$  of 72
**Level 3** $\frac{3}{4}$  of 2 hours (in minutes)

## Extension · Stretch yourself

### Worked example

After spending  $\frac{1}{4}$  of his money and then  $\frac{1}{3}$  of what was left, Ali has \$30. How much did he start with?

1. After  $\frac{1}{4}$ :  $\frac{3}{4}$  left
2. Then  $\frac{2}{3}$  of  $\frac{3}{4} = \frac{1}{2}$  of the money is left
3.  $\frac{1}{2} = \$30$

Answer: \$60

### YOU DO

On your own: read the worked example, then try these.

- E1  $\frac{2}{5}$  of a number is 18. What is  $\frac{3}{4}$  of the number?

- E2 A tank is  $\frac{3}{8}$  full. After adding 60 L it is  $\frac{3}{4}$  full. How much does it hold?

## Lesson summary

Missed the lesson? Start here.

**Remember** 1. “of” means multiply.  
3. Multiply by the numerator.

2. Divide by the denominator.  
4. Change units first if needed.

### Level 1

$\frac{3}{7}$  of 21

1.  $21 \div 7 = 3$
2.  $3 \times 3$

Answer: 9

### Level 2

$\frac{4}{5}$  of 2 km in m

1.  $2 \text{ km} = 2000 \text{ m}$
2.  $2000 \div 5 \times 4$

Answer: 1600 m

### Level 3

$\frac{3}{4}$  of 3 h in min

1.  $180 \div 4 \times 3$

Answer: 135 min

### Extension

Ali's money

1.  $\frac{1}{2}$  left is \$30

Answer: \$60

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LESSON  
4.06

## Multiplying fractions

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- multiply fractions and mixed numerals

## Success criteria

I can:

- ☐ multiply numerators and multiply denominators
- ☐ cancel common factors first
- ☐ change mixed numerals to improper fractions first

**Key terms** Write the definition as your teacher explains it.**Product****Cancel**

Notes

## Level 1 · Getting started

## WE DO

With your teacher: copy what they write on the board.

## Example 1

$\frac{2}{3} \times \frac{1}{5}$

## Example 2

$10 \times \frac{3}{5}$

## YOU DO

On your own: Set A. Evaluate.

1  $\frac{2}{3} \times \frac{1}{5}$

4  $\frac{2}{5} \times \frac{2}{3}$

2  $\frac{1}{3} \times \frac{1}{7}$

5  $10 \times \frac{3}{5}$

3  $\frac{3}{4} \times \frac{1}{2}$

6  $\frac{5}{6} \times 12$

**YOU DO**

On your own: Set B. Evaluate. Cancel first.

①  $\frac{4}{7} \times \frac{5}{8}$

④  $\frac{9}{16} \times \frac{2}{45}$

②  $\frac{3}{10} \times \frac{5}{9}$

⑤  $\frac{4}{15} \times \frac{5}{8}$

③  $\frac{6}{7} \times \frac{7}{12}$

⑥  $\frac{2}{3} \times \frac{9}{10}$

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

**Example 3**

$\frac{4}{7} \times \frac{5}{8}$

**Example 4**

$\frac{9}{16} \times \frac{2}{45}$

**YOU DO**

On your own: Set C. Evaluate.

①  $1\frac{1}{2} \times \frac{1}{2}$

③  $1\frac{1}{2} \times 1\frac{1}{2}$

⑤  $\frac{3}{4} \times 2\frac{1}{3}$

②  $2\frac{1}{5} \times \frac{1}{2}$

④  $2\frac{2}{3} \times \frac{1}{2}$

⑥  $3\frac{3}{5} \times \frac{3}{8}$



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.

**Example 5**  $1\frac{1}{2} \times \frac{1}{2}$

**Example 6**  $2\frac{1}{3} \times 4\frac{2}{3}$

**YOU DO** On your own: Set D. Evaluate.

**1**  $2\frac{1}{3} \times 4\frac{2}{3}$

**3**  $2\frac{1}{2} \times 1\frac{2}{5} \times \frac{1}{3}$

**2**  $3\frac{2}{4} \times \frac{2}{3}$

**4** A recipe needs  $2\frac{3}{4}$  cups of flour. How much for  $\frac{2}{3}$  of the recipe?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

$\frac{3}{5} \times \frac{1}{4}$

**Level 2**

$\frac{5}{9} \times \frac{3}{10}$

**Level 3**

$1\frac{1}{4} \times 2\frac{2}{3}$

## Extension · Stretch yourself

### Worked example

Evaluate  $\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}$ .

1. Cancel: 3, 4 and 5 cancel
2. Leaves  $\frac{2}{6}$

Answer:  $\frac{1}{3}$

### YOU DO

On your own: read the worked example, then try these.

E1 Evaluate  $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{9}{10}$ .

E2 A rectangle is  $3\frac{1}{2}$  m by  $2\frac{2}{3}$  m. Find its area.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Change mixed numerals to improper fractions.
  2. Cancel common factors (top with bottom).
  3. Multiply tops, multiply bottoms.
  4. Write as a simplified mixed numeral.

### Level 1

$$\frac{2}{3} \times \frac{1}{5}$$

1.  $2 \times 1$  and  $3 \times 5$

Answer:  $\frac{2}{15}$

### Level 2

$$\frac{4}{7} \times \frac{5}{8}$$

1. Cancel 4 and 8

Answer:  $\frac{5}{14}$

### Level 3

$$2\frac{1}{3} \times 4\frac{2}{3}$$

1.  $\frac{7}{3} \times \frac{14}{3}$

Answer:  $10\frac{8}{9}$

### Extension

Telescoping product

1. Cancel

Answer:  $\frac{1}{3}$

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LESSON  
4.07

## Dividing fractions

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- divide fractions and mixed numerals

## Success criteria

I can:

- ☐ find the **reciprocal** of a fraction
- ☐ divide by multiplying by the reciprocal
- ☐ change mixed numerals to improper fractions first

**Key terms** Write the definition as your teacher explains it.**Reciprocal****Quotient**

Notes

## Level 1 · Getting started

**WE DO** With your teacher: copy what they write on the board.**Example 1** Write the reciprocal of  $\frac{3}{4}$ , 5 and  $2\frac{1}{2}$ .**Example 2**  $10 \div \frac{1}{5}$ **YOU DO** On your own: Set A. Write the reciprocal.

1  $\frac{3}{4}$

2  $\frac{2}{5}$

3 7

4  $\frac{1}{9}$

5  $\frac{8}{3}$

6  $1\frac{1}{2}$

**YOU DO**

On your own: Set B. Evaluate.

1  $10 \div \frac{1}{5}$

4  $6 \div \frac{2}{3}$

2  $\frac{1}{2} \div 3$

5  $\frac{4}{5} \div 2$

3  $\frac{1}{2} \div \frac{1}{3}$

6  $3 \div \frac{1}{4}$

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

**Example 3**

$\frac{1}{2} \div 3$

**Example 4**

$\frac{3}{4} \div \frac{3}{8}$

**YOU DO**

On your own: Set C. Evaluate.

1  $\frac{3}{4} \div \frac{3}{8}$

3  $\frac{4}{7} \div \frac{5}{8}$

5  $\frac{2}{3} \div \frac{5}{6}$

2  $\frac{3}{5} \div \frac{1}{10}$

4  $\frac{17}{20} \div \frac{3}{5}$

6  $\frac{5}{9} \div \frac{2}{3}$



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.

**Example 5**  $2\frac{1}{4} \div 1\frac{1}{8}$

**Example 6**  $1\frac{7}{8} \div 1\frac{1}{4}$

**YOU DO** On your own: Set D. Evaluate.

**1**  $1\frac{1}{2} \div \frac{1}{2}$

**3**  $1\frac{7}{8} \div 1\frac{1}{4}$

**2**  $2\frac{1}{4} \div 1\frac{1}{8}$

**4** How many  $\frac{3}{4}$  m pieces can be cut from a  $7\frac{1}{2}$  m rope?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**Write the reciprocal of  $\frac{5}{8}$ .
**Level 2**

$\frac{2}{3} \div \frac{1}{6}$

**Level 3**

$3\frac{1}{2} \div 1\frac{3}{4}$

 **Extension · Stretch yourself**
 **Worked example**

A jug holds  $2\frac{1}{4}$  L. How many  $\frac{3}{10}$  L glasses can be filled?

$$1. \frac{9}{4} \div \frac{3}{10} = \frac{9}{4} \times \frac{10}{3}$$

$$2. = \frac{90}{12} = 7\frac{1}{2}$$

**Answer:** 7 full glasses

**YOU DO**

On your own: read the worked example, then try these.

**E1**  $\frac{2}{3}$  of a number is  $\frac{4}{9}$ . Find the number.

**E2** Evaluate  $(\frac{1}{2} + \frac{1}{3}) \div (\frac{1}{2} - \frac{1}{3})$ .

**Lesson summary**

Missed the lesson? Start here.

**Remember** 1. Change mixed numerals to improper fractions. 2. Keep the first fraction.  
3. Change  $\div$  to  $\times$  and flip the second fraction. 4. Multiply and simplify.

 **Level 1**

$$10 \div \frac{1}{5}$$

$$1. 10 \times 5$$

**Answer:** 50

 **Level 2**

$$\frac{3}{4} \div \frac{3}{8}$$

$$1. \frac{3}{4} \times \frac{8}{3}$$

**Answer:** 2

 **Level 3**

$$2\frac{1}{4} \div 1\frac{1}{8}$$

$$1. \frac{9}{4} \times \frac{8}{9}$$

**Answer:** 2

 **Extension**

Glasses from a jug

$$1. 7\frac{1}{2}$$

**Answer:** 7 glasses

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
4.08

## Fraction problems

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- solve problems involving fractions

## Success criteria

I can:

- ☐ decide which operation the problem needs
- ☐ find the “rest” by subtracting fractions from 1
- ☐ check the answer makes sense

**Key terms** Write the definition as your teacher explains it.**Remainder****Total**

Notes

## Level 1 · Getting started

**WE DO** With your teacher: copy what they write on the board.

**Example 1** A punch has 2 L ginger ale,  $1\frac{1}{2}$  L orange juice,  $2\frac{3}{4}$  L lemonade and  $\frac{1}{3}$  L raspberry juice. How many litres of punch?

**Example 2** How much more lemonade than orange juice is in the punch?

**YOU DO** On your own: Set A. Solve.

- 1 Sam ate  $\frac{1}{4}$  of a cake and Ben ate  $\frac{3}{8}$ . How much did they eat together?

- 3  $\frac{3}{5}$  of 40 students walk to school. How many walk?

- 5 A pizza has 8 slices. Lee eats 3. What fraction is left?

- 2 A tank is  $\frac{5}{6}$  full.  $\frac{1}{3}$  is used. How full is it now?

- 4 What fraction of an hour is 20 minutes?

- 6 How many  $\frac{1}{4}$ -hour lessons fit in 3 hours?

**YOU DO** On your own: Set B. Solve.

- ① A ribbon is  $4\frac{1}{2}$  m.  $1\frac{2}{3}$  m is cut off. How much is left?

- ② A bag of flour weighs  $\frac{3}{4}$  kg. What do 6 bags weigh?

- ③  $\frac{1}{3}$  of the class have a dog and  $\frac{1}{4}$  have a cat. What fraction have neither (no one has both)?

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.

- Example 3** A farmer has 24 goats.  $\frac{1}{4}$  are black and  $\frac{2}{3}$  are white. What fraction are neither?

- Example 4** The rest of the goats are split equally between brown and spotted. How many are spotted?

**YOU DO** On your own: Set C. Solve.

- ① A punch uses  $1\frac{1}{2}$  L juice,  $2\frac{3}{4}$  L soda and  $\frac{1}{3}$  L syrup. Find the total.

- ③ A 60 L tank is  $\frac{3}{5}$  full. How many litres to fill it?

- ⑤ How many  $\frac{2}{3}$ -cup serves are in 8 cups?

- ② 24 goats:  $\frac{1}{4}$  black,  $\frac{2}{3}$  white, the rest split equally brown and spotted. How many spotted?

- ④ Mia spends  $\frac{1}{3}$  of \$90, then  $\frac{1}{2}$  of what is left. How much is left?

- ⑥ A car uses  $\frac{3}{4}$  of a tank for 360 km. How far on a full tank?



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Jo reads  $\frac{2}{5}$  of a book on Monday and  $\frac{1}{3}$  on Tuesday. What fraction is left?
**Example 6** The book has 240 pages. How many pages are left?
**YOU DO** On your own: Set D. Harder problems.**1**  $\frac{3}{8}$  of a school's 480 students catch a bus and  $\frac{1}{3}$  of the rest walk. How many walk?
**3** After giving away  $\frac{2}{7}$  of her cards, Zoe has 45 left. How many did she start with?
**2** A rope is cut into pieces of  $\frac{2}{5}$  m. There are 12 pieces and  $\frac{1}{10}$  m left over. How long was the rope?
**4** Is  $\frac{2}{3}$  of  $\frac{3}{4}$  of 80 bigger than  $\frac{1}{2}$  of 80? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**What is  $\frac{2}{5} + \frac{1}{3}$  of a whole?
**Level 2** $\frac{3}{4}$  of 36 students passed. How many failed?
**Level 3**After spending  $\frac{3}{5}$  of his money, Kai has \$28. How much did he start with?

## Extension · Stretch yourself

### Worked example

A cyclist rides  $\frac{1}{3}$  of a trip on day 1 and  $\frac{1}{4}$  of the trip on day 2. She has 50 km left. How long is the trip?

- $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$
- $\frac{5}{12}$  of the trip is 50 km
- $\frac{1}{12}$  is 10 km

Answer: 120 km

### YOU DO

On your own: read the worked example, then try these.

- E1  $\frac{1}{4}$  of a class plays soccer,  $\frac{1}{3}$  plays netball and the other 10 play neither. How many in the class?

- E2 A bottle is  $\frac{2}{3}$  full. After pouring out 150 mL it is  $\frac{1}{6}$  full. How much does it hold?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read the problem twice. Underline the numbers.
  3. "Left" or "the rest" → subtract from 1 (the whole).

2. "Of" → multiply. "Altogether" → add.
4. Answer the question with units.

### Level 1

Punch total

$$1. 2 + 1\frac{1}{2} + 2\frac{3}{4} + \frac{1}{3}$$

Answer:  $6\frac{7}{12}$  L

### Level 2

Goats: neither black nor white

$$1. 1 - \frac{1}{4} - \frac{2}{3}$$

Answer:  $\frac{1}{12}$

### Level 3

Spotted goats

- $\frac{1}{12}$  of 24 = 2
- $2 \div 2$

Answer: 1 goat

### Extension

Cyclist

$$1. \frac{5}{12} = 50 \text{ km}$$

Answer: 120 km

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LESSON  
4.09

## Fractions on a calculator

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- use a calculator to work with fractions

## Success criteria

I can:

- ☐ enter fractions and mixed numerals with the fraction key
- ☐ convert between improper fractions, mixed numerals and decimals
- ☐ check calculator answers with an estimate

**Key terms** Write the definition as your teacher explains it.

Fraction key

S  $\leftrightarrow$  D key

Notes

## Level 1 · Getting started

**WE DO** With your teacher: copy what they write on the board.**Example 1** Enter  $\frac{3}{4} + \frac{1}{6}$ .**Example 2** Enter  $3\frac{1}{2} \times \frac{2}{5}$ .**YOU DO** On your own: Set A. Use a calculator.

1  $\frac{3}{4} + \frac{1}{6}$

2  $\frac{5}{6} + \frac{4}{5}$

3  $\frac{7}{9} - \frac{1}{4}$

4  $\frac{13}{18} - \frac{3}{8}$

5  $2\frac{3}{4} + 1\frac{5}{6}$

6  $5\frac{1}{2} - 2\frac{2}{3}$

**YOU DO** On your own: Set B. Use a calculator.

1  $\frac{9}{16} \times \frac{2}{45}$

4  $2\frac{3}{4} \div 1\frac{1}{5}$

2  $\frac{17}{20} \div \frac{3}{5}$

5  $\frac{5}{8}$  of 360

3  $2\frac{1}{3} \times 4\frac{2}{3}$

6  $\frac{7}{12} \times 48$

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Convert  $\frac{17}{5}$  to a mixed numeral and to a decimal.**Example 4** Evaluate  $\frac{13}{18} - \frac{3}{8}$ .**YOU DO** On your own: Set C. Convert with a calculator.

1  $\frac{17}{5}$  to a mixed numeral

3  $2\frac{3}{8}$  to an improper fraction

5 0.625 to a fraction

2  $\frac{45}{8}$  to a mixed numeral

4  $\frac{7}{20}$  to a decimal

6  $\frac{5}{16}$  to a decimal



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Evaluate  $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$ .**Example 6** Evaluate  $2\frac{1}{4}^2$ .
**YOU DO** On your own: Set D. Use a calculator.

**1**  $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$

**3**  $(\frac{3}{5} + \frac{1}{4}) \times 2\frac{2}{3}$

**2**  $2\frac{1}{4}^2 + \frac{1}{2}$

**4** Estimate  $3\frac{7}{8} \times 2\frac{1}{10}$  first, then use a calculator.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

$\frac{2}{7} + \frac{3}{5}$

**Level 2**

$3\frac{4}{5} \times \frac{5}{6}$

**Level 3**

$2\frac{5}{6} \div \frac{2}{3} - \frac{1}{4}$

## Extension · Stretch yourself

### Worked example

Find the fraction halfway between  $\frac{5}{9}$  and  $\frac{7}{11}$ .

1. Add and divide by 2

2.  $(\frac{5}{9} + \frac{7}{11}) \div 2$

Answer:  $\frac{59}{99}$

### YOU DO

On your own: read the worked example, then try these.

E1 Evaluate  $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$ . What do you notice?

E2 Which is closer to 1:  $\frac{11}{13}$  or  $\frac{13}{15}$ ?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Use the fraction key to enter a fraction.
  2. Use SHIFT for a mixed numeral.
  3. S  $\leftrightarrow$  D switches between fraction and decimal.
  4. Estimate first to check the answer.

### Level 1

$$\frac{3}{4} + \frac{1}{6}$$

1. Enter with the fraction key

Answer:  $\frac{11}{12}$

### Level 2

$$\frac{17}{5}$$

1. Convert to mixed

Answer:  $3\frac{2}{5} = 3.4$

### Level 3

$$4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$$

1. Use brackets if unsure

Answer:  $12\frac{1}{4}$

### Extension

Halfway

1. Add and halve

Answer:  $\frac{59}{99}$

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LESSON  
4.10

## Percentages, fractions and decimals

Date \_\_\_\_\_

**Learning intentions**

We are learning to:

- convert between percentages, fractions and decimals

**Success criteria**

I can:

- ☐ write a percentage as a fraction out of 100 and simplify
- ☐ write a percentage as a decimal by dividing by 100
- ☐ write a fraction or decimal as a percentage by multiplying by 100%

**Key terms** Write the definition as your teacher explains it.**Percentage****Per cent**

Notes

**Level 1 · Getting started****WE DO** With your teacher: copy what they write on the board.**Example 1** Write 13% as a fraction and a decimal.**Example 2** Write 8% as a fraction and a decimal.**YOU DO** On your own: Set A. Write each percentage as a fraction in simplest form.

1 13%

2 8%

3 50%

4 75%

5 35%

6 120%

**YOU DO** On your own: Set B. Write each percentage as a decimal.

1 13%

4 150%

2 8%

5 2.5%

3 45%

6  $7\frac{1}{2}\%$ **Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Write  $7\frac{1}{2}\%$  as a fraction and a decimal.**Example 4** Write  $\frac{7}{10}$  as a percentage.**YOU DO** On your own: Set C. Write each as a percentage.1  $\frac{7}{10}$ 

4 0.6

2  $\frac{23}{25}$ 

5 0.045

3  $\frac{3}{4}$ 

6 1.3



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Write  $\frac{23}{25}$  as a percentage.
**Example 6** Write 0.045 as a percentage.
**YOU DO** On your own: Set D. Complete the conversions.**1** Write  $\frac{3}{8}$  as a decimal and a percentage.
**3** Write  $12\frac{1}{2}\%$  as a fraction.
**2** Write  $\frac{1}{3}$  as a percentage.
**4** Order from smallest to largest: 0.4, 38%,  $\frac{2}{5}$ ,  $\frac{3}{8}$ .

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Write 25% as a fraction.

**Level 2**

Write 0.07 as a percentage.

**Level 3**Write  $\frac{3}{16}$  as a percentage.

## Extension · Stretch yourself

### Worked example

Write  $0.\overline{3}$  ( $0.333\dots$ ) as a percentage and as a fraction.

1.  $0.333\dots \times 100\% = 33.3\dots\%$
2.  $= 33\frac{1}{3}\%$
3. As a fraction:  $\frac{1}{3}$

Answer:  $33\frac{1}{3}\% = \frac{1}{3}$

### YOU DO

On your own: read the worked example, then try these.

- E1 Write  $\frac{2}{3}$  as a percentage.

- E2 Write  $12\frac{1}{3}\%$  as a fraction in simplest form.

## Lesson summary

Missed the lesson? Start here.

**Remember** 1. %  $\rightarrow$  fraction: write over 100, simplify.  
3. Fraction or decimal  $\rightarrow$  %:  $\times 100\%$ .

2. %  $\rightarrow$  decimal:  $\div 100$  (move 2 places left).  
4. Learn:  $\frac{1}{2} = 50\%$ ,  $\frac{1}{4} = 25\%$ ,  $\frac{1}{5} = 20\%$ ,  $\frac{1}{10} = 10\%$ .

### Level 1

13%

1.  $\frac{13}{100}$

Answer:  $\frac{13}{100} = 0.13$

### Level 2

$7\frac{1}{2}\%$

1.  $\frac{15}{2} \div 100 = \frac{15}{200}$

Answer:  $\frac{3}{40} = 0.075$

### Level 3

$\frac{23}{25}$

1.  $\frac{92}{100}$

Answer: 92%

### Extension

$\frac{1}{3}$

1.  $\times 100\% = \frac{100}{3}\%$

Answer:  $33\frac{1}{3}\%$

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LESSON  
4.11

## Percentage of a quantity

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find a percentage of a quantity

## Success criteria

I can:

- ☐ change the percentage to a fraction or decimal and multiply
- ☐ use 10% and 1% to find percentages mentally
- ☐ solve problems with percentages

**Key terms** Write the definition as your teacher explains it.**Discount****Percentage**

Notes

## Level 1 · Getting started

**WE DO** With your teacher: copy what they write on the board.**Example 1** Find 10% of 350.**Example 2** Find 25% of 84.**YOU DO** On your own: Set A. Find mentally.

1 10% of 350

2 50% of 86

3 25% of 84

4 1% of 900

5 20% of 45

6 5% of 60

**YOU DO** On your own: Set B. Evaluate.

1 15% of \$60

4 75% of 36

2 7% of 1200

5 2% of 450

3 30% of 250

6 150% of 40

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Find 15% of \$60.**Example 4** Find 7% of 1200.**YOU DO** On your own: Set C. Evaluate.

1 12.5% of 480 kg

3 8% of 2.5 km (m)

5 17% of 300

2 35% of \$180

4 45% of 3 h (min)

6 0.5% of \$2000



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Find 12.5% of 480 kg.
**Example 6** A \$250 jacket is 30% off. Find the sale price.
**YOU DO** On your own: Set D. Percentage problems.**1** A \$250 jacket is 30% off. Find the sale price.
**3** A \$40 meal has a 15% tip added. Find the total.
**2** 64% of 25 students caught a bus. How many did not?
**4** Which is more: 30% of 70 or 70% of 30? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO**

## Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find 10% of 80.

**Level 2**

Find 35% of 200.

**Level 3**

A \$60 shirt is 15% off. Find the sale price.

## Extension · Stretch yourself

### Worked example

A price rises by 10%, then falls by 10%. Is it back to the original price?

1. Try  $\$100: + 10\% = \$110$
2.  $- 10\%$  of  $\$110 = \$110 - \$11 = \$99$

**Answer:** No: it is 1% lower

### YOU DO

On your own: read the worked example, then try these.

- E1** A \$80 item has 20% off, then another 10% off the new price. Find the final price.

- E2** Is 20% off then 10% off the same as 30% off? Explain.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Change % to a fraction or decimal.
  2. Multiply by the quantity.
  3.  $10\%: \div 10$ .  $1\%: \div 100$ . Build others from these.
  4. Discount: subtract from the price.

### Level 1

10% of 350

1.  $350 \div 10$

**Answer:** 35

### Level 2

15% of \$60

1.  $10\% = \$6$ ,  $5\% = \$3$

**Answer:** \$9

### Level 3

30% off \$250

1.  $30\%$  of  $250 = 75$
2.  $250 - 75$

**Answer:** \$175

### Extension

+10% then -10%

1.  $100 \rightarrow 110 \rightarrow 99$

**Answer:** No, 1% less

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
4.12

## Expressing as a fraction or percentage

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- express one quantity as a fraction or percentage of another

## Success criteria

I can:

- ☐ write the part over the whole
- ☐ convert both quantities to the same unit first
- ☐ multiply by 100% to get a percentage

**Key terms** Write the definition as your teacher explains it.**Part****Whole**

Notes

## Level 1 · Getting started

## WE DO

With your teacher: copy what they write on the board.

## Example 1

Write 51 out of 60 as a fraction and a percentage.

## Example 2

Write 18 out of 24 as a percentage.

## YOU DO

On your own: Set A. Write the first quantity as a fraction of the second, in simplest form.

1 6 out of 10

2 15 out of 20

3 51 out of 60

4 8 out of 12

5 40 out of 100

6 9 out of 45

**YOU DO** On your own: Set B. Write as a percentage.

1 7 out of 10

4 13 out of 20

2 18 out of 24

5 45 out of 50

3 51 out of 60

6 3 out of 8

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Write 300 m as a fraction of 1.5 km.**Example 4** Write 300 m as a percentage of 1.5 km.**YOU DO** On your own: Set C. Change to the same unit, then write as a fraction and a percentage.

1 300 m of 1.5 km

3 25 cm of 2 m

5 \$1.20 of \$4

2 45 min of 2 h

4 400 g of 5 kg

6 750 mL of 3 L



## Level 3 · Challenge

Finished Set D? Try the extension on the next page.

**WE DO** With your teacher: copy what they write on the board.**Example 5** Write 45 min as a percentage of 2 hours.
**Example 6** Mia scored 34/40 and Jake 42/50. Who did better?
**YOU DO** On your own: Set D. Problems.**1** Mia scored 34/40 and Jake scored 42/50. Who did better?
**3** A \$120 price dropped to \$90. What is the percentage decrease?
**2** A team won 18 of 24 games. What percentage did they lose?
**4** A class of 28 has 12 boys. What percentage are girls? (1 d.p.)

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Write 12 out of 16 as a fraction.

**Level 2**

Write 21 out of 25 as a percentage.

**Level 3**

Write 40 cm as a percentage of 2 m.

## Extension · Stretch yourself

### Worked example

A shirt that cost \$40 now costs \$46. Find the percentage increase.

1. Increase = \$6
2.  $\frac{6}{40} \times 100\%$

Answer: 15%

### YOU DO

On your own: read the worked example, then try these.

- E1** A town grew from 2500 to 3100 people. Find the percentage increase.

- E2** A \$75 game is on sale for \$60. What percentage discount is this?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write **part ÷ whole** as a fraction.
  2. Use the same units for both.
  3. Simplify the fraction.
  4. Multiply by 100% for a percentage.

### Level 1

51 out of 60

1.  $\frac{51}{60}$
2.  $\times 100\%$

Answer: 85%

### Level 2

300 m of 1.5 km

1. 1.5 km = 1500 m
2.  $\frac{300}{1500}$

Answer:  $\frac{1}{5} = 20\%$

### Level 3

45 min of 2 h

1.  $\frac{45}{120}$

Answer:  $\frac{3}{8} = 37.5\%$

### Extension

\$40 → \$46

1.  $\frac{6}{40} \times 100\%$

Answer: 15%

This space is left blank on purpose. It is the back of your exit ticket.